

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and fine-tuned **LLM pipelines** for production use cases, integrating **LangChain** for prompt orchestration and managing model inference endpoints on **AWS Lambda**.
- Engineered **NLP workflows** for text classification and entity recognition using **spaCy** and **NLTK**, processing over 50,000 customer feedback entries weekly to identify key themes.
- Collaborated with data scientists to productionize **ML models**, writing **Python** scripts for data validation and creating containerized inference services deployed via **Docker**.
- Designed and implemented scalable **backend services** and **REST APIs** in **Python** (FastAPI) to support high-throughput data ingestion, handling approximately 4,000 requests per second during peak loads.
- Trained and fine-tuned **Hugging Face Transformer models** (BERT, RoBERTa) for downstream NLP tasks, achieving an F1 score of **0.82** on a sentiment analysis benchmark dataset.
- Managed production scoring pipelines on **AWS**, focusing on reliability and performance, participating in on-call rotations for incident response and system observability using **CloudWatch**.
- Contributed to system design discussions and code reviews for new AI features, ensuring adherence to coding standards and performance best practices across the engineering team.

SKILLS

AI/ML – LLM, NLP, RAG, Prompt Engineering, Fine-tuning, Hugging Face Transformers, BERT, RoBERTa, spaCy, NLTK, scikit-learn, PyTorch, TensorFlow, LangChain, LlamaIndex

Programming Languages – Python, Go, SQL, Java

Cloud & DevOps – AWS, Lambda, EC2, S3, CloudWatch, Docker, Kubernetes, Jenkins, Git, Terraform

Backend & API – FastAPI, Flask, REST APIs, gRPC, Microservices, System Design

Data & Databases – PostgreSQL, MySQL, Oracle, Data Modelling, Vector Databases (pgvector)

MLOps & Orchestration – Model Deployment, CI/CD, Observability, LLMops

PROJECTS

AI-Powered Customer Interaction Analysis Platform (HCL Technologies)

- Developed a **Python**-based pipeline to ingest and process customer interaction data from multiple sources, preparing it for downstream **NLP** analysis.
- Built and deployed a **FastAPI** service for real-time analysis of customer feedback, classifying sentiment and extracting key entities with a **Transformer model**, achieving **91% accuracy** on test sets.
- Integrated sentiment scores and entity data into an operational dashboard, enabling quality assurance teams to quickly identify areas for agent coaching and process improvement.

LLM-Enhanced Data Summarization Service (MSc Applied Project)

- Engineered a proof-of-concept **LLM pipeline** using **LangChain** to summarize lengthy technical documents, demonstrating the feasibility of automated knowledge extraction.
- Implemented **retrieval-augmented generation (RAG)** by integrating a vector database (**pgvector**) with an LLM to provide context-aware summaries, improving relevance by an estimated **30%** over baseline summarization.
- Designed the API contract for the summarization service and conducted performance testing, achieving a p95 latency of **450ms** for document processing.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)